OFFICIAL COLLEGE PROJECT ABSTRACT · PROJECT 20 OF 55

Weapon detection using machine learning yolov8

Domain: Computer Vision & Public Security | **Module:** project_20_weapon_detection

OFFICIAL ABSTRACT

Weapon detection using machine learning, specifically through the use of YOLOv8 (You Only Look Once version 8), aims to enhance security systems by automatically identifying weapons in real-time from images. YOLOv8, a state-of-the-art object detection algorithm, leverages deep learning techniques to detect and classify objects with high accuracy and efficiency. By training on large datasets of weapon images, YOLOv8 can identify various types of weapons, such as guns, knives, and explosives, even in complex and crowded environments. This automated detection system can significantly improve public safety by quickly identifying potential threats and enabling faster response times. The use of YOLOv8 for weapon detection provides a scalable, real-time solution that can be integrated into surveillance systems, airports, public venues, and other high-risk areas, enhancing security measures while minimizing human error.

📌 Project Summary & System Specifications

Project Title	Weapon detection using machine learning yolov8
Specialization / Category	Computer Vision & Public Security
Target Technology Stack	Python (Flask), PyTorch / TensorFlow / Scikit-Learn, Modern Web UI
Default Backend Port	`:5020`
Local Implementation	`project_20_weapon_detection`

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